



CUBAN NATIONAL CHEMISTRY CONTEST 2018

ANSWERS TO THE PROBLEMS

“The future of our country must necessarily be a future of men of sciences, must be a future of men of thinking, because that is precisely what we are creating...”

Fidel Castro



Ministerio de Educación
República de Cuba



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Distribution of questions by grade and test sets

	Day 1 (common exam)	Day 2 (individual by grade)
10mo	Problems 1, 2, 3 y 4	Problems 5, 6 y 7
11no	Problems 1, 2, 3 y 4	Problems 8, 9 y 10
12mo	Problems 1, 2, 3 y 4	Problems 10 y 11

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Problem 1. Miscellanea (20 points)

1.1	...	Total	Total
1	idem	20	20
1	---	20	20

Select the correct answer in each case.

1.1. When the following substances are ordered according to growing lattice energy, which is the correct order? MgO, NaF, CaS

A) MgO<NaF<CaS	B) CaS<NaF<MgO	C) NaF<CaS<MgO	D) CaS<MgO<NaF
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1.2. Which of the following oxides is the best reductant reagent? CO₂, NO₂, SiO₂, SO₂

A) NO ₂	B) SiO ₂	C) CO ₂	D) SO ₂
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1.3. Among the molecular substances S₈, H₂O, NH₃, P₄, ¿which one has the highest melting point?

A) NH ₃	B) H ₂ O	C) S ₈	D) P ₄
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1.4. The combustion of methylamine, CH₃NH₂, originates CO₂, N₂ y H₂O. ¿What amount of O₂ is required for the complete reaction of 1,0 mol of methylamine?

A) 2,5	B) 2,25	C) 4,5	D) 3,0
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1.5. The reaction $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{N}_2(\text{g})$ follows the rate law $v = k[\text{NO}]^2[\text{H}_2]$. Which are the units of k if the time is given in seconds and the concentration in mole per liter?

A) L·s·mol ⁻¹	B) mol·L ⁻¹ ·s ⁻¹	C) L ² ·mol ⁻² ·s ⁻¹	D) mol ² ·L ⁻² ·s ⁻¹
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1.6. A manganese oxide contains 30,4% mass of O. Which is the empirical formula?

A) MnO ₂	B) Mn ₃ O ₄	C) Mn ₂ O ₃	D) MnO
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1.7. A temperature rise generally increases the reaction rate in general. Which of the following properties allows us to explain the previous fact?

A) pressure	B) concentration	C) energy of collision	D) contact surface
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1.8. How many unpaired electrons have a chromium atom?

A) 3	B) 4	C) 5	D) 6
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1.9. Which atom has the highest second ionization energy?

A) Al	B) Mg	C) K	D) Na
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1.10. Which periodic properties diminish in group 1 of the periodic table when increasing atomic number? 1. electronegativity, 2. atomic radius, 3. electron affinity

A) only 1	B) 1 and 2	C) 1 and 3	D) 2 and 3
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1.11. How many isomers have the molecular formula C₄H₁₀O?

A) 4	B) 5	C) 6	D) 7
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1.12. What is the molecular geometry of the azide ion, N₃⁻?

A) angular	B) lineal	C) T-shape	D) triangular
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1.13. Which of the following species is polar? 1. BrF₃, 2. CF₄, 3. SbF₅, 4. SF₆

A) only 1	B) only 3	C) 1 and 3	D) 3 and 4
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1.14. In an exothermic reaction is valid that

A) Q = 0	B) ΔT < 0	C) ΔH < 0	D) ΔE < 0
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1.15. When the following species are ordered according to increasing N–O bond distance, which is the correct order? NO_2^- , NO_2^+ , NO_3^- .

A) $\text{NO}_2^- < \text{NO}_2^+ < \text{NO}_3^-$ B) $\text{NO}_3^- < \text{NO}_2^- < \text{NO}_2^+$ C) $\text{NO}_2^+ < \text{NO}_3^- < \text{NO}_2^-$ **D) $\text{NO}_2^+ < \text{NO}_2^- < \text{NO}_3^-$**

1.16. What is the hybridization of the carbon atoms in propyne?

A) sp^2 B) sp^3 **C) sp , sp^3** D) sp , sp^2 , sp^3

1.17. When metals Mg, Zn and Pb are ordered according to growing reactivity with non-oxidizing acids, which is the correct order?

A) $\text{Mg} < \text{Zn} < \text{Pb}$ **B) $\text{Pb} < \text{Zn} < \text{Mg}$** C) $\text{Pb} < \text{Mg} < \text{Zn}$ D) $\text{Mg} < \text{Pb} < \text{Zn}$

1.18. Which of the following acids has a notable acid character? 1. CrO_3 , 2. MnO_2 , 3. SO_2 , 4. P_2O_3

A) 2, 3, 4 B) 1, 2, 3 C) 1, 2, 4 **D) 1, 3, 4**

1.19. Which of the following binary substances doesn't have a chemical bonding with strong ionic character? 1. CsAu, 2. MgO, 3. GaAs, 4. CaC_2

A) 1 B) 2 **C) 3** D) 4

1.20. Which of the following metals doesn't form an oxide when burnt in the presence of dioxygen? 1. K, 2. Be, 3. Ca, 4. Li

A) 1 B) 2 C) 3 D) 4

Problem 2. Identification of substances (10 points)

2.1	2.2	Total	Total
8	12	20	10
8	12	20	10

A student received the assignment to identify the substances constituting two solid samples (mixture A and B). It is known that sample A may contain SiO_2 , Na, NiCl_2 , CrCl_3 , CoCl_2 or $\text{Ba}(\text{NO}_3)_2$; while sample B may be formed by AgNO_3 , Ag, $\text{Cu}(\text{NO}_3)_2$, KI or Fe. The student reacted each of the samples with common inorganic reagents and obtained the following results:

Mixture A: It is green. It dissolves completely in water originating a green solution and there is no observable evidence of any strong exothermic reaction. After the addition of sodium hydroxide to the resultant solution, a green precipitate is obtained which partially dissolves in excess of the reagent forming an intensely colored green solution. The solid portion is dissolved when adding hydrochloric acid. The addition of sulfuric acid to the original solution doesn't form any precipitate.

Mixture B: It is grey. It dissolves partially in water originating a colorless solution. After several minutes of standing, the solid is inert to hydrochloric acid or sodium hydroxide. On the contrary, when the colorless solution is treated with sodium hydroxide, a white precipitate forms, which is completely soluble in excess of the reagent.

2.1. Find the substances composing mixtures A and B.

Mixture A: NiCl_2 and CrCl_3	4 marks
Mixture B: Zn and AgNO_3	4 marks

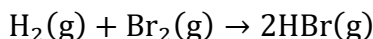
2.2. Write the equations corresponding to the chemical reactions that happened during the identification.

$\text{NiCl}_2(\text{ac}) + 2\text{NaOH}(\text{ac}) \rightarrow \text{Ni}(\text{OH})_2(\text{s}) + 2\text{NaCl}(\text{ac})$	2 marks
$\text{CrCl}_3(\text{ac}) + 3\text{NaOH}(\text{ac}) \rightarrow \text{Cr}(\text{OH})_3(\text{s}) + 3\text{NaCl}(\text{ac})$	2 marks
$\text{Cr}(\text{OH})_3(\text{s}) + \text{NaOH}(\text{ac}) \rightarrow \text{Na}[\text{Cr}(\text{OH})_4](\text{ac})$	1 mark
$\text{Ni}(\text{OH})_2(\text{s}) + 2\text{HCl}(\text{ac}) \rightarrow \text{NiCl}_2(\text{ac}) + \text{H}_2\text{O}(\text{l})$	1 mark
$\text{Zn}(\text{s}) + 2\text{AgNO}_3(\text{ac}) \rightarrow \text{Zn}(\text{NO}_3)_2(\text{ac}) + 2\text{Ag}(\text{s})$	2 marks
$\text{Zn}(\text{NO}_3)_2 + 2\text{NaOH}(\text{ac}) \rightarrow \text{Zn}(\text{OH})_2(\text{s}) + 2\text{NaNO}_3(\text{ac})$	2 marks
$\text{Zn}(\text{OH})_2(\text{s}) + 2\text{NaOH}(\text{ac}) \rightarrow \text{Na}_2[\text{Zn}(\text{OH})_4](\text{ac})$	2 marks

Problem 3. Kinetics (10 points)

3.1	3.2	3.3	3.4	Total	Total
5	5	4	6	20	10
5	5	4	6	20	10

From a kinetics point of view, many chemical reactions can be classified as chain reactions. In chain reactions, an intermediate formed in one step generates another intermediate in the following step and so on. These intermediates are named propagators. An example of such a process is the reaction between hydrogen and bromine.



The following mechanism has been proposed to explain the results of the kinetics studies.

Initiation	$\text{Br}_2 \rightarrow 2\text{Br}$	$v = k_1[\text{Br}_2]$
Propagation	$\text{Br} + \text{H}_2 \rightarrow \text{HBr} + \text{H}$	$v = k_2[\text{Br}][\text{H}_2]$
	$\text{H} + \text{Br}_2 \rightarrow \text{HBr} + \text{Br}$	$v = k_3[\text{H}][\text{Br}_2]$
Retardation	$\text{H} + \text{HBr} \rightarrow \text{H}_2 + \text{Br}$	$v = k_4[\text{H}][\text{HBr}]$
Termination	$2\text{Br} \rightarrow \text{Br}_2$	$v = k_5[\text{Br}]^2$

3.1. Identify reactants, products and intermediates. State the molecularity of the last step of the reaction. Which is the rate-limiting step?

Reactants: Br_2, H_2	Products: HBr	Intermediates: H, Br	5 marks
Molecularity: 2	Rate limiting step: Initiation		

3.2. Write down mathematical expressions relating the variation of the concentration of the intermediates in time as a function of the rate constants of individual steps and the concentrations of the species involved.

$$\frac{d[\text{Br}]}{dt} = 2k_1[\text{Br}_2] + k_3[\text{H}][\text{Br}_2] + k_4[\text{H}][\text{HBr}] - k_2[\text{Br}][\text{H}_2] - 2k_5[\text{Br}]^2 \quad 3 \text{ marks}$$

$$\frac{d[\text{H}]}{dt} = k_2[\text{Br}][\text{H}_2] - k_3[\text{H}][\text{Br}_2] - k_4[\text{H}][\text{HBr}] \quad 2 \text{ marks}$$

The experimental rate law for the previous reaction is a complex expression (see below); however, under certain experimental conditions, the expression is simplified and a pseudo-order behavior is observed.

$$\text{Rate law: } \frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]}$$

3.3. Which are the partial reaction orders with respect to Br_2 at the exact moment the reaction starts and when it is close to its end? Justify your answer.

When the reaction starts, there are only reactants,

$$\frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2]} = k[\text{H}_2][\text{Br}_2]^{1/2} \quad 1 \text{ mark}$$

The partial order of Br_2 is $1/2$. 1 mark

When the reaction is about to finish, there are almost no reactants,

$$\frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{k'[\text{HBr}]} \quad 1 \text{ mark}$$

The partial order of Br_2 is $3/2$. 1 mark

3.4. Find mathematical expressions for the total activation energy and pre-exponential factor, assuming that the reaction rate is $k = 2k_2(k_1/k_5)^{1/2}$.

$$k = 2k_2 \sqrt{\frac{k_1}{k_5}} = 2A_2 e^{-\frac{E_{a2}}{RT}} \sqrt{\frac{A_1 e^{-\frac{E_{a1}}{RT}}}{A_5 e^{-\frac{E_{a5}}{RT}}}} \quad 1 \text{ mark}$$

$$k = 2A_2 \sqrt{\frac{A_1}{A_5}} e^{-\frac{E_{a2}}{RT}} \sqrt{\frac{e^{-\frac{E_{a1}}{RT}}}{e^{-\frac{E_{a5}}{RT}}}} = 2A_2 \sqrt{\frac{A_1}{A_5}} e^{-\frac{E_{a2} + \frac{E_{a1}}{2} - \frac{E_{a5}}{2}}{RT}} \quad 1 \text{ mark}$$

$$A_{\text{Total}} = 2A_2 \left(\frac{A_1}{A_5} \right)^{1/2} \quad 2 \text{ marks}$$

$$E_{a\text{Total}} = E_{a2} + \frac{E_{a1}}{2} - \frac{E_{a5}}{2} \quad 2 \text{ marks}$$

Problem 4. Stoichiometry I (10 points)

3.1	3.2	Total	Total
4	16	20	10

A sample containing copper, zinc and aluminum of mass 3,000 g reacted with an excess of hydrochloric acid and 1,643 L (measured at 25,00 °C and 101,3 kPa) of gas, a solution and a solid were obtained. The solid residue was filtered, dried and weighed, having a mass equal to 0,850 g.

4.1. Write the equations corresponding to the chemical reactions that happened.



4.2. Find the composition (in mass percent) of the initial sample.

$$\omega(\text{Cu}) = \frac{m(\text{Cu})}{m(\text{total})} \cdot 100 = \frac{0,850}{3,00} \cdot 100 = 28,3\% \quad 3 \text{ marks}$$

$$3,000 = m(\text{Cu}) + m(\text{Zn}) + m(\text{Al})$$

$$2,150 = m(\text{Zn}) + m(\text{Al}) \quad (1) \quad 2 \text{ marks}$$

$$n(\text{H}_2) = \frac{pV}{RT} = \frac{101,3 \cdot 1,643}{8,3145 \cdot 298,15} = 0,06714 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{H}_2) = n(\text{Zn}) + \frac{3}{2}n(\text{Al}) = \frac{m(\text{Zn})}{A_r(\text{Zn})} + \frac{3}{2} \frac{m(\text{Al})}{A_r(\text{Al})} \quad 1 \text{ mark}$$

$$0,06714 = \frac{1}{65,39}m(\text{Zn}) + \frac{3}{2} \cdot \frac{1}{26,98}m(\text{Al}) \quad (2) \quad 3 \text{ marks}$$

After solving the system formed by equations 1 and 2 one gets

$$m(\text{Zn}) = 1,300 \text{ g} \quad 2 \text{ marks}$$

$$\omega(\text{Zn}) = \frac{m(\text{Zn})}{m(\text{total})} \cdot 100 = \frac{1,30}{3,00} \cdot 100 = 43,3\% \quad 1 \text{ mark}$$

$$m(\text{Al}) = 0,850 \text{ g} \quad 2 \text{ marks}$$

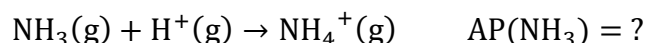
$$\omega(\text{Al}) = \frac{m(\text{Al})}{m(\text{total})} \cdot 100 = \frac{0,850}{3,00} \cdot 100 = 28,3\% \quad 1 \text{ mark}$$

Problem 5. Thermochemistry (20 points)

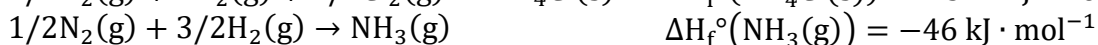
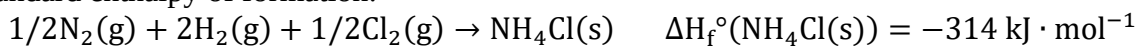
5.1	5.2	5.3	5.4	5.5	5.6	5.7	Total	Total
10	3	4	8	3	4	8	40	20
10	3	4	8	3	4	8	40	20

Proton affinity (AP) is the energy associated with the process in which a chemical species gains a proton, an H^+ ion. The value of this physical quantity provides quantitative information about the acid-base characteristics.

5.1. Use the data given to calculate the proton affinity of ammonia (NH_3).



Standard enthalpy of formation:



Standard enthalpy of dissociation:



Electron affinity:



Ionization energy:



Lattice energy:



$\text{NH}_3(\text{g}) \rightarrow 1/2\text{N}_2(\text{g}) + 3/2\text{H}_2(\text{g})$	$-\Delta H_f^\circ(\text{NH}_3(\text{g}))$	1 mark
$\text{H}^+(\text{g}) + \text{e}^- \rightarrow \text{H}(\text{g})$	$-\text{I}(\text{H})$	1 mark
$\text{NH}_4\text{Cl}(\text{s}) \rightarrow \text{NH}_4^+(\text{g}) + \text{Cl}^-(\text{g})$	$-\text{U}(\text{NH}_4\text{Cl})$	1 mark
$1/2\text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) + 1/2\text{Cl}_2(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$	$\Delta H_f^\circ(\text{NH}_4\text{Cl}(\text{s}))$	1 mark
$\text{Cl}^-(\text{g}) \rightarrow \text{Cl}(\text{g}) + \text{e}^-$	$-\text{EA}(\text{Cl})$	1 mark
$\text{Cl}(\text{g}) \rightarrow 1/2\text{Cl}_2(\text{g})$	$-1/2\Delta H_{\text{dis}}^\circ(\text{Cl}_2)$	1 mark
$\text{H}(\text{g}) \rightarrow 1/2\text{H}_2(\text{g})$	$-1/2\Delta H_{\text{dis}}^\circ(\text{H}_2)$	1 mark
$\text{NH}_3(\text{g}) + \text{H}^+(\text{g}) \rightarrow \text{NH}_4^+(\text{g})$	$\text{AP}(\text{NH}_3) = ?$	
$\text{AP} = -\Delta H_f^\circ(\text{NH}_3(\text{g})) - \text{I}(\text{H}) - \text{U}(\text{NH}_4\text{Cl}) + \Delta H_f^\circ(\text{NH}_4\text{Cl}(\text{s})) - \text{EA}(\text{Cl})$		
$\quad - 1/2\Delta H_{\text{dis}}^\circ(\text{Cl}_2) - 1/2\Delta H_{\text{dis}}^\circ(\text{H}_2)$		
$= 46 - 1313 + 651 - 314 + 349 - 242/2 - 431/2 = -918 \text{ kJ} \cdot \text{mol}^{-1}$		
		3 marks

5.2. A substance is a good base when it can easily accept a proton (ion H^+). Proton affinities of phosphine (PH_3) and ammonia are $-785 \text{ kJ}\cdot\text{mol}^{-1}$ and $-854 \text{ kJ}\cdot\text{mol}^{-1}$, respectively. Which of the two is the best base?

Ammonia is a better base because of its higher proton affinity.

3 marks

5.3. Phosphine and ammonia molecules are isoelectronic, but HPH and HNH bond angles have an amplitude of $93,5^\circ$ and $107,8^\circ$, respectively. Which atomic orbitals are involved to a more significant extent in the formation of P-H and N-H bonds?

P – H bond: $1s(\text{H})$ and $3p(\text{P})$

2 marks

N – H bond: $1s(\text{H})$ and $2sp_3(\text{N})$

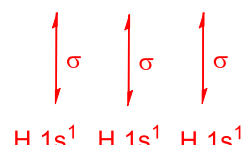
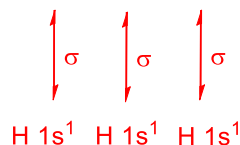
2 marks

5.4. Represent the electronic configuration of ammonia and phosphine molecules according to the Valence Bond Theory. Only use the orbitals in the valence shell for your analysis.

N(hib): $[\text{He}] 2sp_3^2 2sp_3^1 2sp_3^1 2sp_3^1$

P: $[\text{Ne}] 3s^2 3p^1 3p^1 3p^1$

2 marks



2 marks

The enthalpy of a covalent bond A-B, where A and B are two atoms, is the average energy needed to break one mole of these bonds at constant pressure. In diatomic molecules, the bond enthalpy is numerically equal to the dissociation enthalpy.

5.5. The standard mean bond enthalpies for P-H and N-H bonds are $322 \text{ kJ}\cdot\text{mol}^{-1}$ and $388 \text{ kJ}\cdot\text{mol}^{-1}$, respectively. Justify why the N-H bond is stronger.

N atom is in period 2, therefore its orbitals have a size similar to H atom

$1s$ orbitals, which allows for stronger interaction.

3 marks

5.6. The strongest chemical bond known to date is that of the dinitrogen molecule. Explain this experimental fact.

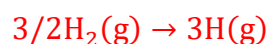
En la molécula de N_2 existe un enlace triple y el tamaño de los orbitales es pequeño; ambos factores provocan que el enlace sea muy fuerte.

4 marks

5.7. Calculate the standard enthalpy of dissociation of the dinitrogen molecule. Elaborate on an appropriate thermodynamic cycle and use the data from the problem.



1 mark



1 mark



2 marks



1 mark

$$\Delta H_f^\circ(\text{NH}_3(\text{g})) = 1/2\Delta H_{\text{dis}}^\circ(\text{N}_2) + 3/2\Delta H_{\text{dis}}^\circ(\text{H}_2) - 3\Delta H^\circ(\text{N} - \text{H})$$

$$-46 = 1/2\Delta H_{\text{dis}}^\circ(\text{N}_2) + 3/2 \cdot 431 - 3 \cdot 388$$

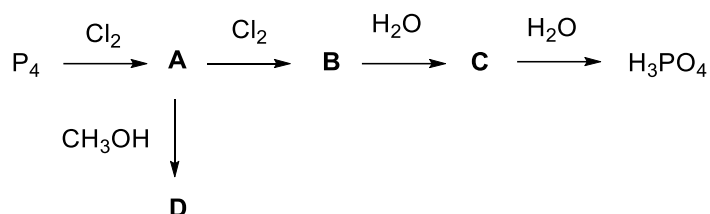
$$\Delta H^\circ(\text{N} - \text{N}) = 943 \text{ kJ} \cdot \text{mol}^{-1}$$

3 marks

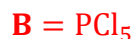
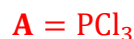
Problem 6. Chemical bonding (15 points)

6.1	6.2	6.3	6.4	6.5	6.6	Total	Total
8	3	2	2	9	6	30	15
8	3	2	2	9	6	30	15

The following diagram contains some of the reactions of tetraphosphorous and its compounds.



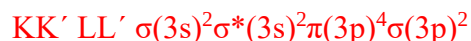
6.1. Find the substances represented by letters A, B, C and D.



8 marks

6.2. Though the most stable allotropic modification of this element is P_4 (in the form of white or red phosphorous), it is possible to detect a diatomic molecule in the gaseous phase. Predict its magnetic properties by using the Molecular Orbital Theory.

The electronic structure of P_2 molecule is



2 marks

All the electrons are paired, therefore the molecule should be diamagnetic.

1 mark

6.3. Phosphorus(V) oxide, P_4O_{10} , is a strong dehydrating reagent that reacts with nitric acid to form metaphosphoric acid, HPO_3 and a white ionic solid. Write down the equation corresponding to this chemical reaction.



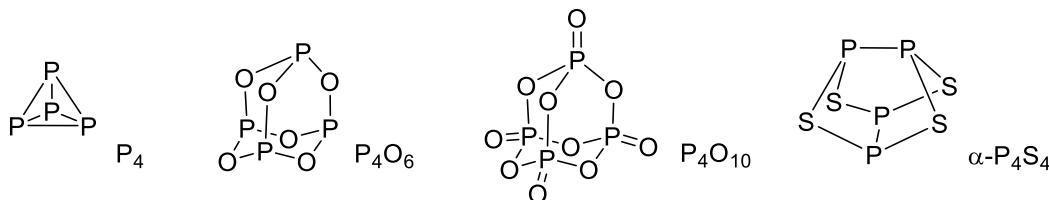
2 marks

6.4. Phosphine, PH_3 , is a highly toxic compound; it can be obtained by disproportionation tetraphosphorous in the presence of potassium hydroxide. Write the chemical equation knowing that potassium dihydrogen hypophosphate, KH_2PO_2 , is also formed.

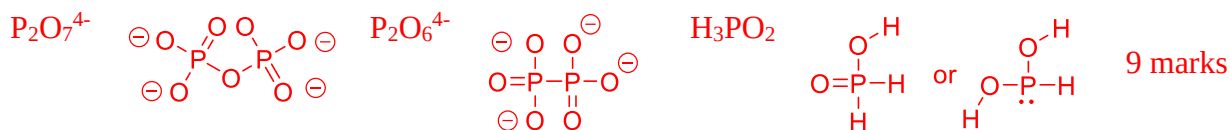


2 marks

Phosphorous element is known for its capacity to form molecules and inorganic ions with great molecular complexity when combined with non-metallic elements. In these substances, it is usual to find cycles with various sizes in which the phosphorous atoms alternate with atoms of another chemical element.

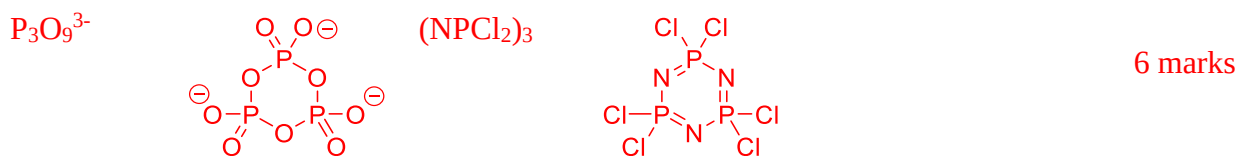


6.5. Draw Lewis structures for the species diphosphate, $\text{P}_2\text{O}_7^{4-}$; hypophosphate, $\text{P}_2\text{O}_6^{4-}$; and hypophosphorous acid, H_3PO_2 .



9 marks

6.6. Suggest Lewis structures for the species $\text{P}_3\text{O}_9^{3-}$ and $(\text{NPCl}_2)_3$ if both molecules are symmetric and contain a six-atom cycle.



6 marks

Problem 7. Stoichiometry II (15 points)

7.1	7.2	7.3	Total	Total
10	10	10	30	15
10	10	10	30	15

It is a common practice to express the formula of a mineral as a mixture of oxides. For example, the formula of kaolin, the main constituent of clay, can be described as $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$. In this problem, your task will be to find the formula (oxide composition) of several important minerals.

Beryl is the most important mineral of beryllium and is a ternary oxide of beryllium, silicon and aluminum. In beryl Be:Al mass ratio is approximately 1:2 and Si:O molar ratio is 1:3.

7.1. Calculate the empirical formula of beryl.

$$\frac{m(\text{Be})}{m(\text{Al})} = \frac{1}{2} = \frac{9,01 \cdot n(\text{Be})}{26,98 \cdot n(\text{Al})} \rightarrow \frac{n(\text{Be})}{n(\text{Al})} \approx \frac{3}{2}$$

4 marks

Supposing berilum formula is $3\text{BeO} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2$

2 marks

$$\frac{n(\text{Si})}{n(\text{O})} = \frac{1}{3} = \frac{x}{2x+6} \rightarrow 3x = 6 + 2x \rightarrow x = 6 \quad 2 \text{ marks}$$

Beryl formula is: $3\text{BeO} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$ 2 marks

Hydroxyapatite is the mineral containing the largest amount of calcium and phosphorous inside the human body; it is part of the bones and the dental enamel. It is formed by hydroxide phosphate and calcium ions. Its composition in mass percent is 39,9% of calcium and 18,5% phosphorous.

7.2. Calculate the empirical formula of hydroxyapatite.

En 100 g de hydroxyapatite, 1 mark

$$n(\text{Ca}^{2+}) = n(\text{Ca}) = \frac{m(\text{Ca})}{A_r(\text{Ca})} = \frac{39,9}{40,08} = 0,996 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{PO}_4^{3-}) = n(\text{P}) = \frac{m(\text{P})}{A_r(\text{P})} = \frac{18,5}{30,97} = 0,597 \text{ mol} \quad 1 \text{ mark}$$

From the charge balance one gets

$$2n(\text{Ca}^{2+}) = 3n(\text{PO}_4^{3-}) + n(\text{OH}^-) \quad 2 \text{ marks}$$

$$n(\text{OH}^-) = 0,201 \text{ mol} \quad 1 \text{ mark}$$

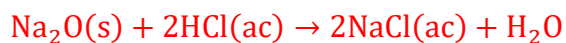
$$n(\text{Ca}^{2+}) : n(\text{PO}_4^{3-}) : n(\text{OH}^-) = 0,996 : 0,597 : 0,20 \approx 5 : 3 : 1 = 10 : 6 : 2 \quad 2 \text{ marks}$$

Hydroxyapatite formula: $10\text{CaO} \cdot 3\text{P}_2\text{O}_5 \cdot \text{H}_2\text{O}$ 2 marks

Borax is used to manufacture of enamel, glass and ceramic; it can be easily transformed into boric acid, which has numerous uses. Borax contains the chemical elements boro, sodium, hydrogen and oxygen; and can be considered a hydrated mix sodium and boron oxide. 1,91 g of borax react with 0,0100 mol of hydrochloric acid, producing 1,25 g of boric acid.

7.3. Calculate the empirical formula of borax. $M(\text{H}_3\text{BO}_3) = 61,84 \text{ g} \cdot \text{mol}^{-1}$

If borax formula is $x\text{Na}_2\text{O} \cdot y\text{B}_2\text{O}_3 \cdot z\text{H}_2\text{O}$



$$n(\text{Na}_2\text{O}) = n(\text{HCl})/2 = 0,0100/2 = 0,00500 \text{ mol} \quad 1 \text{ mark}$$

$$M(\text{Na}_2\text{O}) = 2 \cdot A_r(\text{Na}) + A_r(\text{O}) = 2 \cdot 22,99 + 16,00 = 61,98 \text{ g} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$m(\text{Na}_2\text{O}) = n(\text{Na}_2\text{O}) \cdot M(\text{Na}_2\text{O}) = 0,00500 \cdot 61,98 = 0,309 \text{ g} \quad 1 \text{ mark}$$

$$n(\text{B}_2\text{O}_3) = \frac{n(\text{H}_3\text{BO}_3)}{2} = \frac{m(\text{H}_3\text{BO}_3)}{2M(\text{H}_3\text{BO}_3)} = \frac{1,24}{2 \cdot 61,84} = 0,0101 \text{ mol} \quad 1 \text{ mark}$$

$$M(\text{B}_2\text{O}_3) = 2 \cdot A_r(\text{B}) + 3 \cdot A_r(\text{O}) = 2 \cdot 10,81 + 3 \cdot 16,00 = 69,62 \text{ g} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$m(\text{B}_2\text{O}_3) = n(\text{B}_2\text{O}_3) \cdot M(\text{B}_2\text{O}_3) = 0,0101 \cdot 69,62 = 0,703 \text{ g} \quad 1 \text{ mark}$$

$$m(\text{H}_2\text{O}) = m(\text{bórax}) - m(\text{Na}_2\text{O}) - m(\text{B}_2\text{O}_3) = 1,91 - 0,309 - 0,703 \approx 0,90 \text{ g} \quad 1 \text{ mark}$$

$$M(\text{H}_2\text{O}) = 2 \cdot A_r(\text{H}) + A_r(\text{O}) = 2 \cdot 1,01 + 16,00 = 18,02 \text{ g} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$n(\text{H}_2\text{O}) = \frac{m(\text{H}_2\text{O})}{M(\text{H}_2\text{O})} = \frac{0,90}{18,02} = 0,0499 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{Na}_2\text{O}):n(\text{B}_2\text{O}_3):n(\text{H}_2\text{O}) = 0,00500:0,0101:0,0499 \approx 1:2:10$$

$$\text{Borax formula: } \text{Na}_2\text{O} \cdot 2\text{B}_2\text{O}_3 \cdot 10\text{H}_2\text{O} \quad 1 \text{ mark}$$

Problem 8. Chemical equilibrium (15 points)

8.1	8.2	8.3	8.4	8.5	8.6	Total	Total
6	4	2	6	8	4	30	15
6	4	2	6	8	4	30	15

Iodine is an element essential for life and can be found in living organisms in small amounts. When heated under atmospheric pressure (101,3 kPa), iodine sublimates, that is, it changes the aggregation state from solid to vapor.



8.1. Calculate the sublimation temperature of iodine.

Species	$\Delta H_f^\circ (\text{kJ} \cdot \text{mol}^{-1})$	$S^\circ (\text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1})$
$\text{I}_2(\text{s})$	0,0	116,7
$\text{I}_2(\text{g})$	62,3	260,7

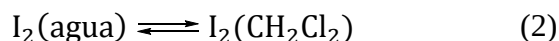
$$\Delta H_r^\circ = \Delta H_f^\circ[\text{I}_2(\text{g})] - \Delta H_f^\circ[\text{I}_2(\text{s})] = 62,3 - 0 = 62,3 \text{ kJ} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$\Delta S_r^\circ = S^\circ[\text{I}_2(\text{g})] - S^\circ[\text{I}_2(\text{s})] = 260,7 - 116,7 = 144 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \quad 1 \text{ mark}$$

$$\Delta G_r^\circ = -RT \ln p_r(\text{I}_2) = -8,314 \cdot 298 \cdot \ln 1,013 = -32,0 \text{ J} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

$$T = \frac{\Delta G_r^\circ - \Delta H_r^\circ}{-\Delta S_r^\circ} = \frac{-32 - 62300}{-144} = 433 \text{ K} = 160 \text{ }^\circ\text{C} \quad 2 \text{ marks}$$

When dichloromethane (CH_2Cl_2) is added to a solution of iodine in water, iodine is distributed among the aqueous and organic phases according to the following equilibrium.



8.2. Calculate the equilibrium constant for the distribution of iodine among water and dichloromethane at 298 K. $\Delta G_f^\circ(\text{kJ} \cdot \text{mol}^{-1})$: $\text{I}_2(\text{ac}) = 16,4$; $\text{I}_2(\text{CH}_2\text{Cl}_2) = 4,00$.

$$\Delta G_r^\circ = \Delta G_f^\circ[\text{I}_2(\text{CH}_2\text{Cl}_2)] - \Delta G_f^\circ[\text{I}_2(\text{ac})]$$

$$\Delta G_r^\circ = 4,00 - 16,4 = -12,4 \text{ kJ} \cdot \text{mol}^{-1} \quad 2 \text{ marks}$$

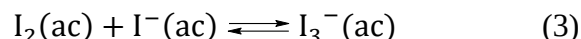
$$K_2 = \exp\left(\frac{\Delta G_r^\circ}{-RT}\right) = \exp\left(\frac{-12400}{-8,314 \cdot 298}\right) = 149 \quad 2 \text{ marks}$$

Note: If you did not solve the previous item, consider the constant of distribution equal to 200 when solving the rest of the problem.

8.3. Why is iodine more soluble in organic substances than in water?

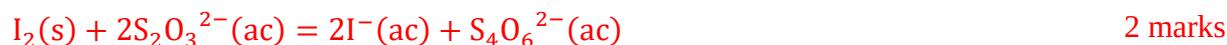
Iodine is an apolar substance, the same as most organic solvents, while water is a polar substance. 2 marks

In an aqueous solution, there is an equilibrium involving the species iodine, iodide and triiodide, according to the following equation.



A suitable method to determine the solution concentrations of I_2 and I_3^- is to titrate with a standard solution of thiosulfate ($\text{S}_2\text{O}_3^{2-}$). When these iodine species interact with thiosulfate, an oxidation-reduction reaction occurs, forming iodide and tetrathionate ($\text{S}_4\text{O}_6^{2-}$).

8.4. Write the equations for the chemical reactions that happen during the titration of I_2 and I_3^- with thiosulfate. Which is the oxidation number of sulfur in thiosulphate ion?



Oxidation numbers of sulfur are 4+ and 0. Answer 2+ is also accepted. 2 marks

Aiming to determine the constant for equilibrium (3), the following experiments were performed at 298 K.

- 50,0 mL of an aqueous solution of KI $0,010 \text{ mol} \cdot \text{L}^{-1}$ were added to 25 mL of I_2 solution in CH_2Cl_2 , forming two separate phases, one aqueous and the other organic.
- Through a spectrophotometric method, the iodine concentration in the organic phase was determined to be $1,20 \cdot 10^{-2} \text{ mol} \cdot \text{L}^{-1}$.
- An aliquot of 25,0 mL from the aqueous phase was taken and mixed with an excess of KI solution to avoid the evaporation of iodine. Then, the resultant solution was titrated with $\text{Na}_2\text{S}_2\text{O}_3$ $0,0100 \text{ mol} \cdot \text{L}^{-1}$ solution. The equivalence point was reached after adding 3,10 mL of the titrant.

8.5. Calculate the equilibrium concentrations of I^- , I_2 y I_3^- in the aqueous phase and the equilibrium constant for the formation of triiodide ion (3) at 298 K.

$$[I_2(ac)] = \frac{[I_2(CH_2Cl_2)]}{K_2} = \frac{1,2 \cdot 10^{-2}}{149} = 8,05 \cdot 10^{-5} \text{ mol} \cdot L^{-1} \quad 1 \text{ mark}$$

$$n[I_2(ac)] + n[I_3^-(ac)] = \frac{1}{2} \cdot n(S_2O_3^{2-}) = \frac{1}{2} \cdot 0,0100 \cdot 3,10 = 1,55 \cdot 10^{-2} \text{ mmol} \quad 2 \text{ marks}$$

$$[I_2(ac)] + [I_3^-(ac)] = \frac{1,55 \cdot 10^{-2}}{25,0} = 6,20 \cdot 10^{-4} \text{ mol} \cdot L^{-1} \quad 1 \text{ mark}$$

$$[I_3^-(ac)] = 6,20 \cdot 10^{-4} - 8,05 \cdot 10^{-5} = 5,40 \cdot 10^{-4} \text{ mol} \cdot L^{-1} \quad 1 \text{ mark}$$

$$[I^-(ac)] = 0,0100 - [I_3^-(ac)] = 0,0100 - 5,40 \cdot 10^{-4} = 9,46 \cdot 10^{-3} \text{ mol} \cdot L^{-1} \quad 1 \text{ mark}$$

$$K_3 = \frac{[I_3^-(ac)]}{[I_2(ac)][I^-(ac)]} = \frac{5,40 \cdot 10^{-4}}{8,05 \cdot 10^{-5} \cdot 9,46 \cdot 10^{-3}} = 709 \quad 2 \text{ marks}$$

It is possible to calculate the constant for equilibrium (3) from electrochemical data. Having the following standard potential electrodes.

$E^\circ(V)$: $I_3^-(ac)/I^-(ac) = 0,536$; $I_2(ac)/I^-(ac) = 0,621$.

8.6. Calculate the equilibrium constant for the formation of triiodide ion (3) at 298 K employing the data of standard electrode potential.

For equilibrium (3) is true that

$$\Delta E_r^\circ = E^\circ(I_2/I^-) - E^\circ(I_3^-/I^-) = 0,621 - 0,536 = 0,085 \text{ V} \quad 2 \text{ marks}$$

$$K_3 = \exp\left(\frac{-zF\Delta E_r^\circ}{-RT}\right) = \exp\left(\frac{-2 \cdot 96485 \cdot 0,085}{-8,314 \cdot 298}\right) = 750 \quad 2 \text{ marks}$$

Problem 9. Titrations (15 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	Total	Total
5	5	2	2	2	3	2	9	30	15
5	5	2	2	2	3	2	9	30	15

Oxalic acid is a substance widely used as a primary standard in acid-base titrations. It is an acid of moderate strength with dissociation constants in water at 25 °C, equal to $K_{a1} = 1,41 \cdot 10^{-2}$ and $K_{a2} = 1,55 \cdot 10^{-4}$.

9.1. Calculate the pH of a solution of $H_2C_2O_4$ 0,0100 mol·L⁻¹.

Considering only the first disotiation process,

$$c(H_2C_2O_4) \approx [H_2C_2O_4] + [HC_2O_4^-] \quad y \quad [H^+] \approx [HC_2O_4^-] \quad 2 \text{ marks}$$

$$K_{a1} = \frac{[H^+][HC_2O_4^-]}{[H_2C_2O_4]} \approx \frac{[H^+]^2}{c(H_2C_2O_4) - [H^+]}$$

$$1,41 \cdot 10^{-2} = \frac{[H^+]^2}{0,01 - [H^+]} \quad 1 \text{ mark}$$

$$[H^+] = 6,76 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$\text{pH} = 2,17 \quad 1 \text{ mark}$$

More accurate calculation renders $\text{pH} = 2,16$ so the approximation is correct.

9.2. Calculate the pH of a KHC_2O_4 $0,0100 \text{ mol} \cdot \text{L}^{-1}$ solution.

Considering hydrogenooxalate as a monoprotic acid,

$$c(\text{HC}_2\text{O}_4^-) \approx [\text{HC}_2\text{O}_4^-] + [\text{C}_2\text{O}_4^{2-}] \quad y \quad [H^+] \approx [\text{C}_2\text{O}_4^{2-}] \quad 2 \text{ marks}$$

$$K_{a2} = \frac{[H^+][\text{C}_2\text{O}_4^{2-}]}{[\text{HC}_2\text{O}_4^-]} \approx \frac{[H^+]^2}{c(\text{HC}_2\text{O}_4^-) - [H^+]}$$

$$1,55 \cdot 10^{-4} = \frac{[H^+]^2}{0,01 - [H^+]} \quad 1 \text{ mark}$$

$$[H^+] = 1,17 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$\text{pH} = 2,93 \quad 1 \text{ mark}$$

Hydrolysis constant is much more smaller than K_{a2} . Therefore the approximate solution considering KHC_2O_4 as a typical ampholyte renders $\text{pH} = 2,83$. Penalize with – 2 marks.

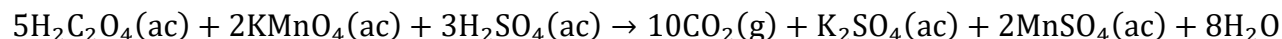
One of the experimental challenges of the participant in the X Central American and VIII Caribbean Chemistry Olympiad (Havana, July 17th-23rd, 2016) consisted in the analysis of a mixture of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ and $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ by a combination of acid-base and redox titrations. In the following, the experiments that could have been conducted by one of the instructors during the preparation for the practical exam are described.

A mixture of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ ($M = 126,08 \text{ g} \cdot \text{mol}^{-1}$) and $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ ($M = 184,43 \text{ g} \cdot \text{mol}^{-1}$) was rinsed into a volumetric flask with distilled water and was completed up to the top to prepare exactly a solution containing 250,0 mL (problem solution).

25,0 mL of the problem solution were taken and transferred into a 100 mL Erlenmeyer flask. A few drops of phenolphthalein ($\text{pK}_a = 9,4$) were added and the mixture was titrated with a solution of $0,100 \text{ mol} \cdot \text{L}^{-1}$ sodium hydroxide from a Mohr burette. The volume consumed from the titrant was 19,0 mL on average.

5,00 mL of the problem solution were taken and transferred to a 100 mL Erlenmeyer flask. 10 mL of $2,6 \text{ mol} \cdot \text{L}^{-1}$ sulfuric acid were added, the mixture was carefully heated and titrated with a $0,00990$

$\text{mol} \cdot \text{L}^{-1}$ potassium permanganate solution from an amber burette. The volume consumed of the titrant was 11,2 mL on average. The reaction that occurs in this case is:



9.3. Write the chemical equation corresponding to the titration with sodium hydroxide.



2 marks

9.4. The solutions of sodium hydroxide and potassium hydroxide employed during the titration were standardized previously to carry out the experiments because they are not primary standards. Mention four characteristics that must have a substance to be considered a primary standard.

A primary standard must have a definite composition and high molecular mass, must be stable and inert towards the air components; it has to react quantitatively and stoichiometric.

2 marks

9.5. How can you detect the end point of each titration?

Acid – base titration: by color change; phenolphthalein is colorless in acid media and red in basic media.

1 mark

Redox titration: by color change; Mn^{2+} ions are colorless while MnO_4^- ions are purple.

1 mark

9.6. Demonstrate that in the conditions where the redox titration is performed, practically all the oxalate of the sample is in the form of oxalic acid. Consider that the pH value is 0,0.

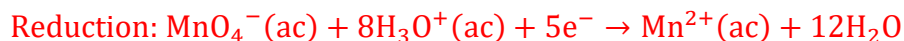
$$\alpha_0 = \frac{[\text{H}_2\text{C}_2\text{O}_4]}{c(\text{H}_2\text{C}_2\text{O}_4)} = \frac{[\text{H}^+]^2}{[\text{H}^+]^2 + K_{a1}[\text{H}^+] + K_{a1}K_{a2}}$$

1 mark

$$\alpha_0 = \frac{1,0^2}{1,0^2 + 1,41 \cdot 10^{-2} \cdot 1,0 + 1,41 \cdot 10^{-2} \cdot 1,55 \cdot 10^{-4}} = 0,986$$

2 marks

9.7. Write the half-reactions of oxidation and reduction corresponding to the redox process.



1 mark



1 mark

9.8. Calculate the composition (in mass percent) of the mixture analyzed by the instructor.

$$2c(\text{H}_2\text{C}_2\text{O}_4)V(\text{aliquot}) = c(\text{NaOH})V(\text{NaOH})$$

1 mark

$$c(\text{H}_2\text{C}_2\text{O}_4) = \frac{0,100 \cdot 19,0}{2 \cdot 25,0} = 0,0380 \text{ mol} \cdot \text{L}^{-1}$$

1 mark

$$2[c(\text{H}_2\text{C}_2\text{O}_4) + c(\text{KHC}_2\text{O}_4)]V(\text{alícuota}) = 5c(\text{KMnO}_4)V(\text{KMnO}_4)$$

1 mark

$$c(\text{H}_2\text{C}_2\text{O}_4) + c(\text{KHC}_2\text{O}_4) = \frac{5 \cdot 0,00990 \cdot 11,2}{2 \cdot 5,00} = 0,05544 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$c(\text{KHC}_2\text{O}_4) = 0,01744 \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$m(\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}) = c \cdot V(\text{total}) \cdot M = 0,0380 \cdot 0,250 \cdot 126,08 = 1,198 \text{ g} \quad 1 \text{ mark}$$

$$m(\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{O}) = c \cdot V(\text{total}) \cdot M = 0,01744 \cdot 0,250 \cdot 184,43 = 0,804 \text{ g} \quad 1 \text{ mark}$$

$$\omega(\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}) = \frac{m(\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O})}{m(\text{total})} \cdot 100 = \frac{1,198}{1,198 + 0,804} \cdot 100 \approx 60\% \quad 1 \text{ mark}$$

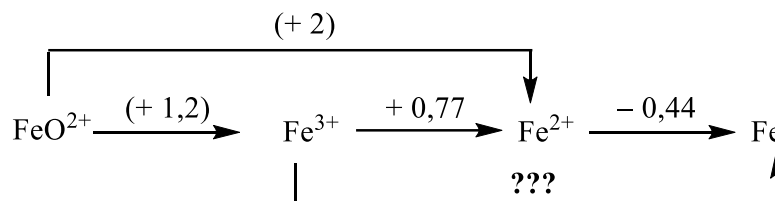
$$\omega(\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{O}) = 100 - 60 = 40\% \quad 1 \text{ mark}$$

Problem 10. Electrochemistry (20 points)

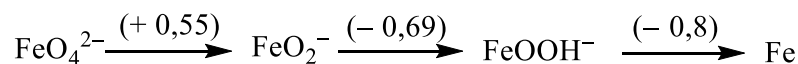
10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
4	4	4	4	6	6	6	6	40	20
4	4	4	4	6	6	6	6	40	20

The following Latimer diagram corresponds to iron species in acid and basic media.

Acid media



Basic media



10.1. In which media is Fe(III) more oxidant? In which media metallic Fe is more reductant?

Iron(III) is more oxidant in acid media. 2 marks

Metallic iron is more reductant in basic media. 2 marks

10.2. Does any of the iron species disproportionate? Justify.

No. For any of the species the reduction potential is higher than the oxidation potential. 2 marks

2 marks

10.3. May species Fe(III) and Fe coexist simultaneously in an acid solution? Justify numerically.



$$\Delta E^\circ = E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) - E^\circ(\text{Fe}^{2+}/\text{Fe}) = 0,77 - (-0,44) = 1,21 \text{ V} \quad 2 \text{ marks}$$

No, the potential difference is positive. 1 mark

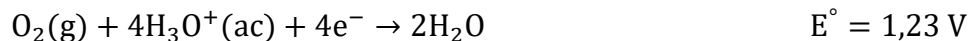
10.4. Calculate the standard potential $E^\circ(\text{Fe}^{3+}/\text{Fe})$.

$$3E^\circ(\text{Fe}^{3+}/\text{Fe}) = E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) + 2E^\circ(\text{Fe}^{2+}/\text{Fe}) \quad 2 \text{ marks}$$

$$E^\circ(\text{Fe}^{3+}/\text{Fe}) = \frac{0,77 + 2 \cdot (-0,44)}{3} = -0,037 \text{ V} \quad 2 \text{ marks}$$

It is possible to understand the redox processes occurring in natural waters with the aid of electrochemical data. For example, in a lake, cold in the depth and warmer on the surface, the temperature gradients avoid vertical mixing. On the surface water is completely oxygenated and iron is present in the form of insoluble particles of $\text{Fe}(\text{OH})_3$. These particles tend to sink. More profound, the content of O_2 is low. If the organic content and other sources of reducing agent are high enough, the hydroxide will be reduced, and the iron will dissolve in the form of Fe^{2+} . Then, the formed ions will diffuse to the surface, where they will be in contact with O_2 and oxidize again to $\text{Fe}(\text{OH})_3$.

10.5. Estimate the approximate potential on the surface of an ordinary lake. Consider that the temperature is 25°C , the pressure is 1,01 bar and the pH is 7,0. Air composition is N_2 78%, O_2 21% and Ar 1% (volume percent).



$$p(\text{O}_2) = \chi(\text{O}_2) \cdot p(\text{total}) = 0,21 \cdot 1,01 = 0,212 \text{ bar} \quad 1 \text{ mark}$$

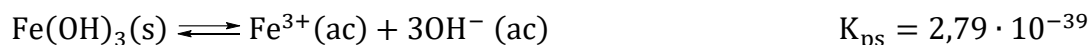
$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-7,0} = 1,00 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

$$E = E^\circ - \frac{RT}{4F} \ln \left(\frac{1}{p_r(\text{O}_2)[\text{H}_3\text{O}^+]^4} \right) \quad 2 \text{ marks}$$

$$E = 1,23 - \frac{8,3145 \cdot 298,15}{4 \cdot 96485} \ln \left(\frac{1}{0,212 \cdot [10^{-7}]^4} \right) = 0,81 \text{ V} \quad 2 \text{ marks}$$

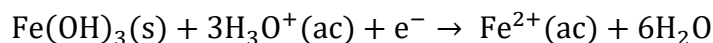
In fact, natural water conditions are slightly different from those calculated by you due to the presence of several species in the solution.

10.6. Calculate Fe^{3+} and Fe^{2+} concentrations on the surface of the lake. Consider that the temperature is 25°C , the pH is 6,5 and the potential is 0,65 V.



$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-6,5} = 3,16 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1}$	1 mark
$[\text{OH}^-] = \frac{K_W}{[\text{H}_3\text{O}^+]} = \frac{1,00 \cdot 10^{-14}}{3,16 \cdot 10^{-7}} = 3,16 \cdot 10^{-8} \text{ mol} \cdot \text{L}^{-1}$	1 mark
$[\text{Fe}^{3+}] = \frac{K_{\text{ps}}}{[\text{OH}^-]^3} = \frac{2,79 \cdot 10^{-39}}{[3,16 \cdot 10^{-8}]^3} = 8,84 \cdot 10^{-17} \text{ mol} \cdot \text{L}^{-1}$	1 mark
$\text{Fe}^{3+}(\text{ac}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{ac}) \quad E^\circ = 0,77 \text{ V}$	1 mark
$E = E^\circ - \frac{RT}{F} \ln \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right) \rightarrow [\text{Fe}^{2+}] = [\text{Fe}^{3+}] \cdot \exp \left(\frac{F(E - E^\circ)}{-RT} \right)$	1 mark
$[\text{Fe}^{2+}] = 8,84 \cdot 10^{-17} \cdot \exp \left(\frac{96485 \cdot (0,65 - 0,77)}{-8,3145 \cdot 298,15} \right) = 9,44 \cdot 10^{-15} \text{ mol} \cdot \text{L}^{-1}$	1 mark

10.7. Demonstrate that the standard potential corresponding to the following solution process that occurs close to the bottom of the lake is $E^\circ = 0,97 \text{ V}$.



$\text{Fe}(\text{OH})_3(\text{s}) \rightarrow \text{Fe}^{3+}(\text{ac}) + 3\text{OH}^-(\text{ac})$	$\Delta_r G_1^\circ = -RT \ln K_{\text{ps}}$	1 mark
$3\text{H}_3\text{O}^+(\text{ac}) + 3\text{OH}^-(\text{ac}) \rightarrow 6\text{H}_2\text{O}$	$\Delta_r G_2^\circ = 3RT \ln K_W$	1 mark
$\text{Fe}^{3+}(\text{ac}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{ac})$	$\Delta_r G_3^\circ = -FE^\circ(\text{Fe}^{3+}/\text{Fe}^{2+})$	1 mark
$\text{Fe}(\text{OH})_3(\text{s}) + 3\text{H}_3\text{O}^+(\text{ac}) + \text{e}^- = \text{Fe}^{2+}(\text{ac}) + 6\text{H}_2\text{O}$	$\Delta_r G_4^\circ = -FE^\circ$	1 mark
$\Delta_r G_4^\circ = \Delta_r G_1^\circ + \Delta_r G_2^\circ + \Delta_r G_3^\circ$		
$-FE^\circ = -RT \ln K_{\text{ps}} + 3RT \ln K_W - FE^\circ(\text{Fe}^{3+}/\text{Fe}^{2+})$		1 mark
$E^\circ = -\frac{-8,314 \cdot 298,15 \cdot \ln 2,79 \cdot 10^{-39} + 3 \cdot 8,314 \cdot 298,15 \cdot \ln 1,00 \cdot 10^{-14} - 96485 \cdot 0,77}{96485}$		
$E^\circ = 0,974 \text{ V}$		1 mark

10.8. Calculate Fe^{3+} and Fe^{2+} concentrations at the bottom of the lake. Consider that the temperature is 15°C , the pH is 6,0 and the potential is 0,15 V. Overlook the effects of the temperature on the equilibrium constant and the standard potential.

$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-6,0} = 1,00 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1}$	1 mark
$[\text{OH}^-] = \frac{K_W}{[\text{H}_3\text{O}^+]} = \frac{1,00 \cdot 10^{-14}}{1,00 \cdot 10^{-6}} = 1,00 \cdot 10^{-8} \text{ mol} \cdot \text{L}^{-1}$	1 mark

$$[\text{Fe}^{3+}] = \frac{K_{ps}}{[\text{OH}^-]^3} = \frac{2,79 \cdot 10^{-39}}{[1,00 \cdot 10^{-8}]^3} = 2,79 \cdot 10^{-15} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$



$$E = E^\circ - \frac{RT}{F} \ln \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right) \rightarrow [\text{Fe}^{2+}] = [\text{Fe}^{3+}] \cdot \exp \left(\frac{F(E - E^\circ)}{-RT} \right) \quad 1 \text{ mark}$$

$$[\text{Fe}^{2+}] = 2,79 \cdot 10^{-15} \cdot \exp \left(\frac{96485 \cdot (0,15 - 0,77)}{-8,3145 \cdot 298,15} \right) = 1,88 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1} \quad 1 \text{ mark}$$

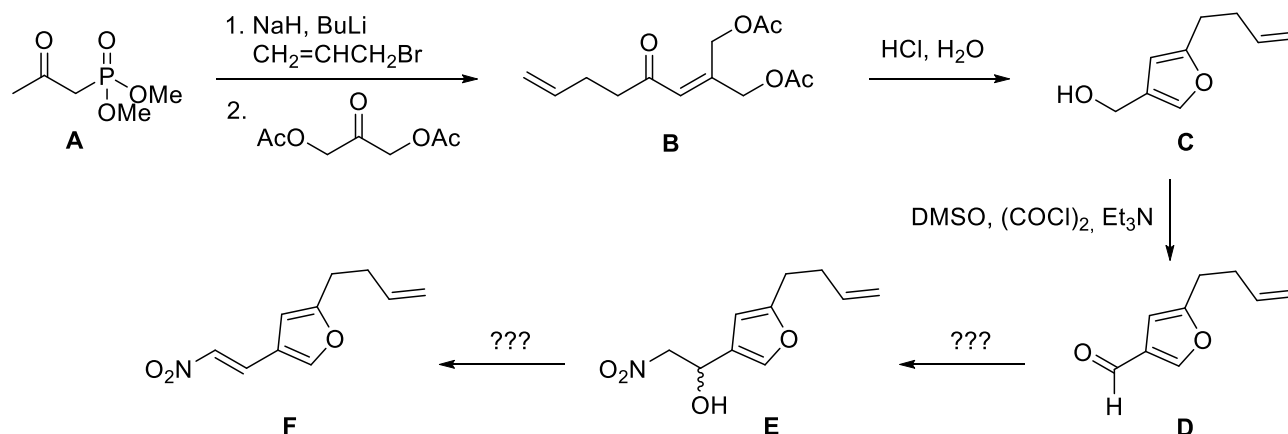
It is also possible to employ the reduction half – reaction from the previous item.

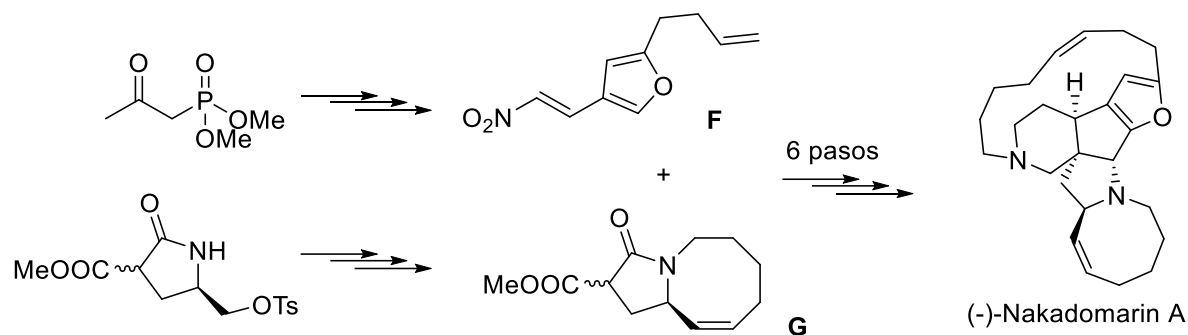
Problem 11. Organic Chemistry (30 points)

10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	10.9	10.10	10.11	10.12	Total	Total
6	8	4	3	4	4	3	3	11	7	4	3	60	30
6	8	4	3	4	4	3	3	11	7	4	3	60	30

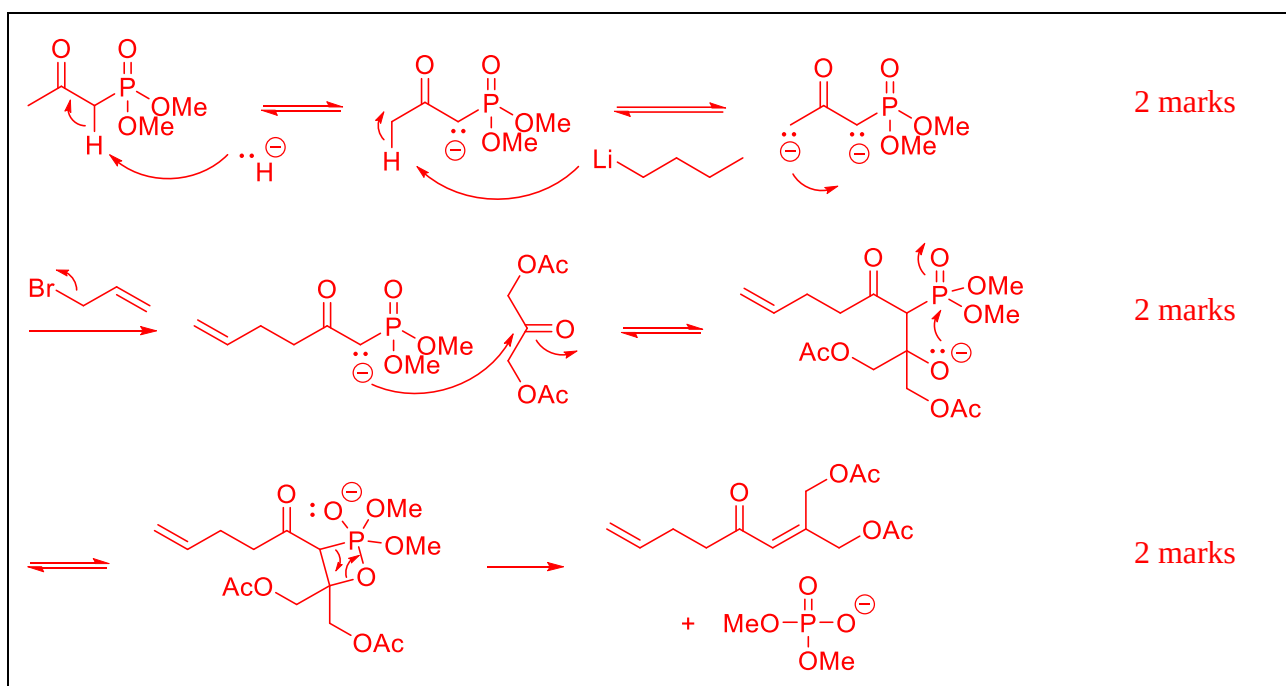
The carbonyl group in aldehydes and ketones is one of the most versatile groups from a synthetic point of view. Therefore, numerous compounds used as intermediaries in complex synthetic routes have this chemical function.

(-)-Nakadomarin A is a member of manzamines, alkaloids isolated from marine sponges, which possess a unique hexacyclic structure. The following scheme shows a fragment of the total synthesis of this alkaloid that was carried out by Darren J. Dixon and coworkers (Jakubec, P.; Cockfield, D. M.; Dixon, D. J., J. Am. Chem. Soc. 2009, 131, 16632–16633).

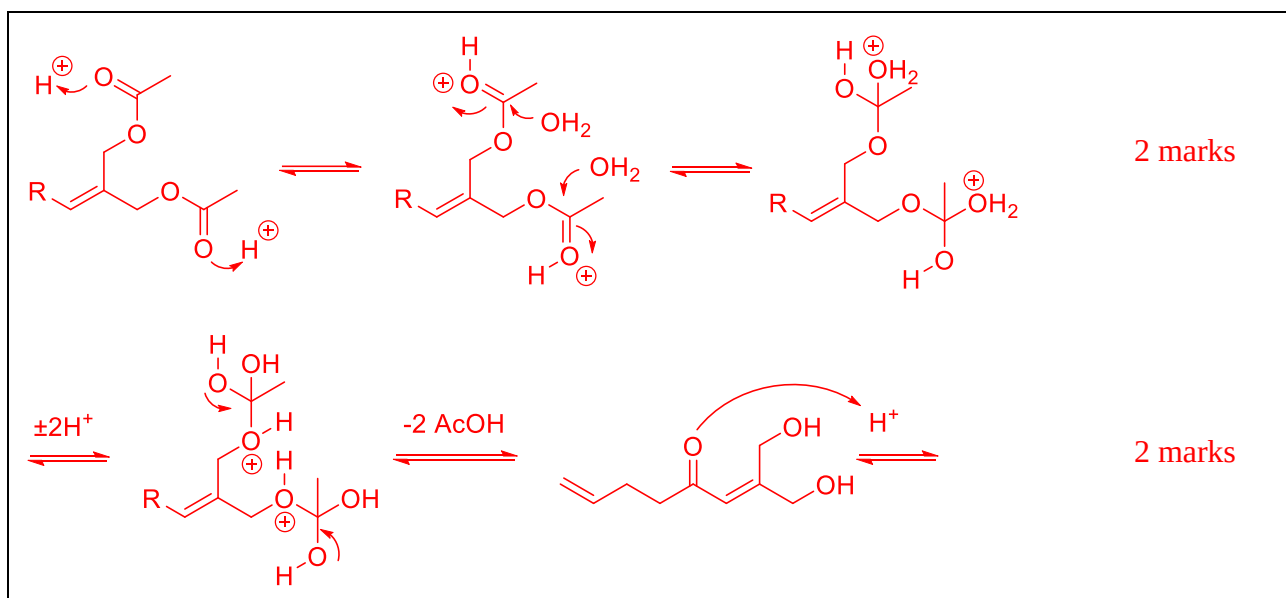


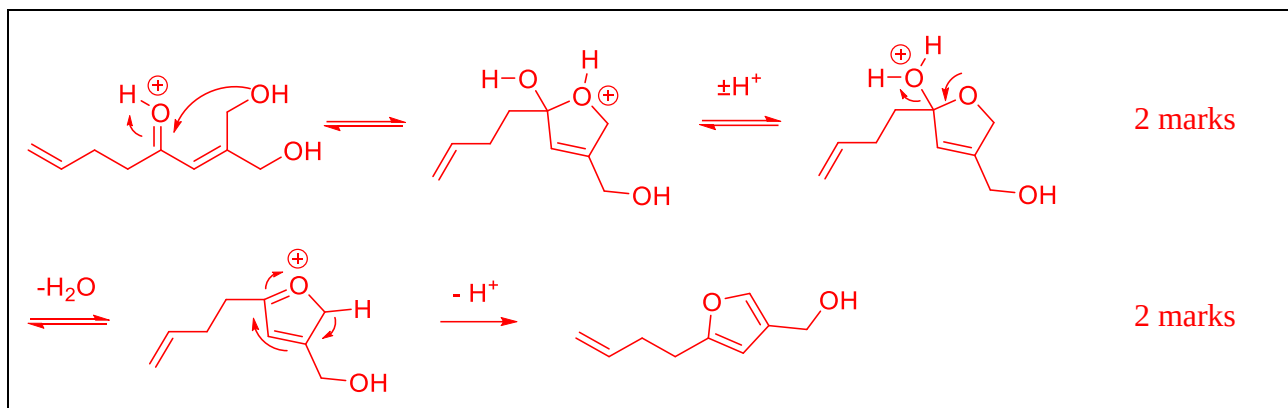


11.1. Propose a mechanism for the preparation of **B** from **A**.



11.2. Propose a mechanism for the preparation of **C** from **B**.





11.3. The obtaining of **C** is favored for forming an aromatic furan ring. Mention three characteristics a system must have to be considered aromatic.

Cyclic structure with conjugated π bonds.

Each atom in the ring has a non-hybrid p orbital that overlaps forming the π system. 4 marks

The structure is flat or quasi flat allowing for an effective overlap.

The delocalization of the π electrons in the ring must diminish the electronic energy.

11.4. What is the function of the mixture DMSO, $(\text{COCl})_2$, Et_3N : oxidizing agent, reducing agent, solvent, dehydrating or deprotection mixture? DMSO = dimethylsulfoxide

Oxidizing reagent

3 marks

11.5. Suggest reagents for the conversion of **D** to **E**.

CH_3NO_2 , KOH

4 marks

Other strong bases such as tBuOK and NaOH are accepted.

11.6. Suggest reagents that allow the conversion of **E** to **F**.

$\text{CH}_3\text{SO}_2\text{Cl}$, $\text{N}(\text{CH}_3)_3$

4 marks

Other sulfur halides as TsCl and other bases are accepted.

11.7. What is the name of the chemical reaction happening when intermediates **F** and **G** are mixed in the presence of a base?

Conjugate 1,4-addition or Michael's addition.

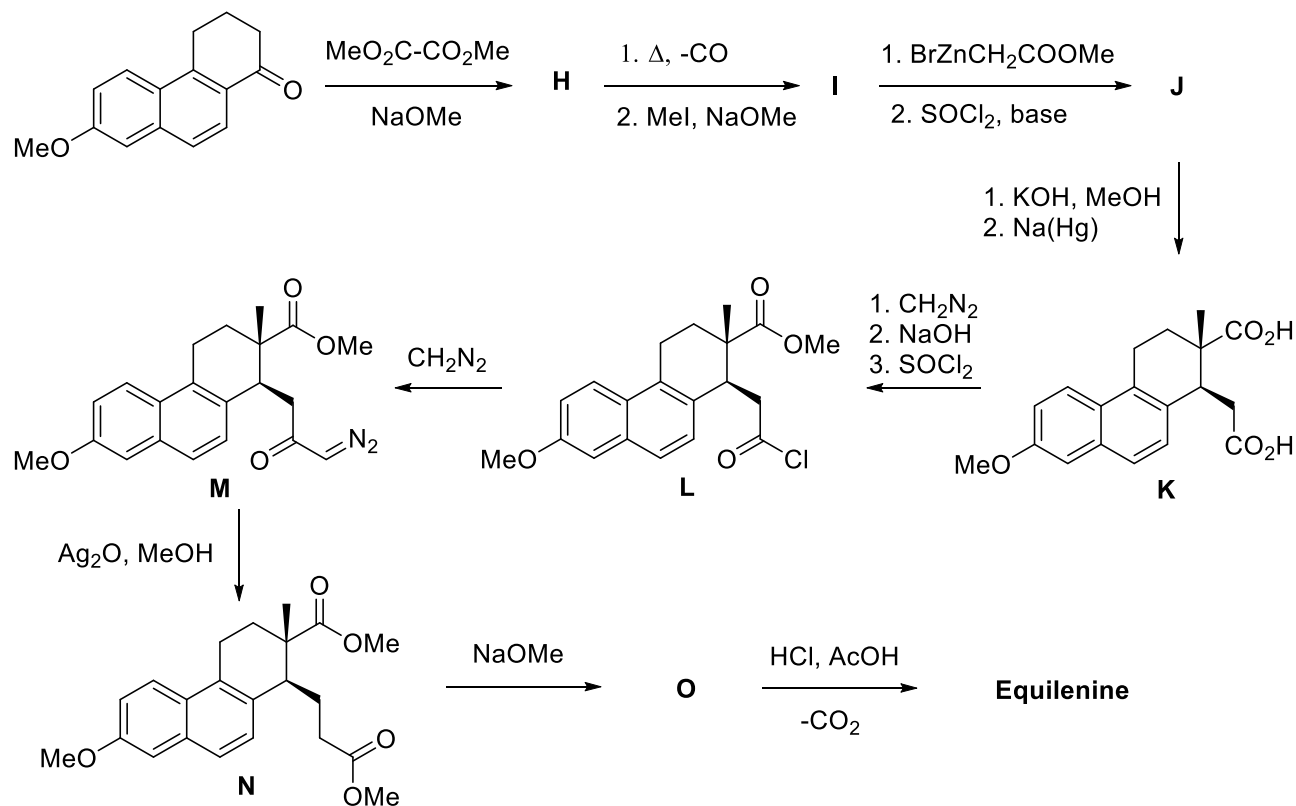
3 marks

11.8. Which is the maximum number of stereoisomers that (-)-Nakadomarin A can have?

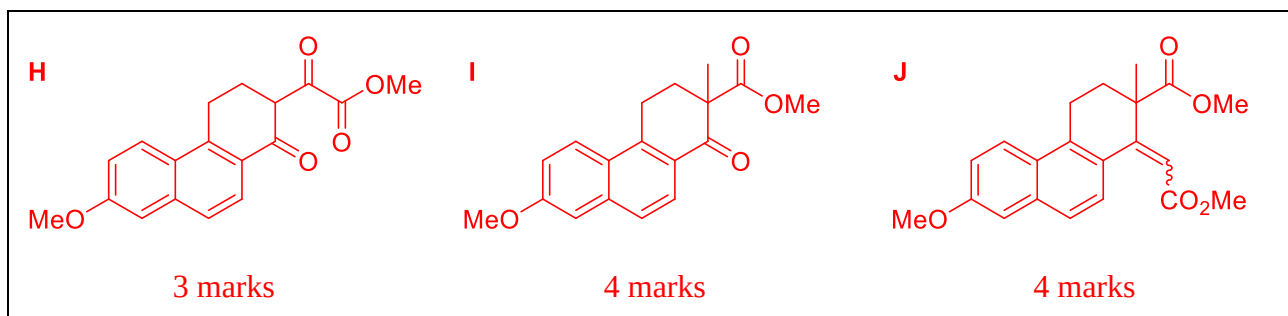
Nakadomarin has four asymmetric centers and two double bonds, hence the maximum number of stereoisomers is $2^6 = 64$.

3 marks

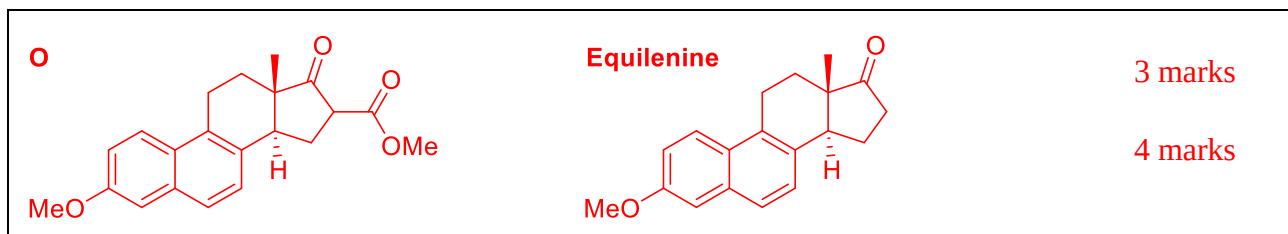
Equilenine is a steroidal sexual hormone from the family of estrogens. It was the first natural product of a complex structure that was completely synthesized. The scheme corresponding to the total synthesis of equilenine (Bachmann, W. E.; Cole, W.; Wilds, A. L., J. Am. Chem. Soc. 1940, 62 (4), 824–839) is given below.



11.9. Represent the structures of compounds **H**, **I** and **J**.



11.10. Represent the structures of compounds **O** and *Equilenine*.



11.11. During the transformation of α -diazoketone **M** catalyzed by Ag(I), a very reactive intermediate **X** is formed, which quickly adds methanol giving **N**; **X** also reacts quickly with water, alcohols and amines. Represent the structure of **X**.



11.12. Diazomethane (CH_2N_2) is a reagent widely employed in organic synthesis. **Suggest** what is the objective of the previously described synthetic step.

Transform a carboxylic acid in the corresponding methyl ester.

1 mark

Extend the carbon chain of a carboxylic acid by one additional atom.

2 marks